

[illegible]

1. Phospholipid ship

WHAT YOU SEE

Pirate ship hull labeled 'THE PHOSPHOLIPID', sailing the sea of dark red blood.

WHAT IT ENCODES

Antiphospholipid syndrome is an acquired autoimmune thrombophilia in which antibodies target phospholipid-BINDING PROTEINS (chiefly β 2-glycoprotein-I bound to anionic phospholipid), activating endothelium, platelets, monocytes, and complement to drive thrombosis in arteries, veins, and the microvasculature. It occurs as primary APS or secondary to SLE (~30–40% of lupus patients carry aPL). It is a leading cause of acquired hypercoagulability and pregnancy morbidity in young women.

BOARD TRAP

WRONG to think APS is a bleeding disorder despite the prolonged aPTT — it is a CLOTTING disorder. The in-vitro coagulation abnormality (lupus anticoagulant) does not translate to bleeding risk in vivo.

2. Skull & crossbones (highest risk)

WHAT YOU SEE

Skull-and-crossbones flag tagged 'HIGHEST RISK'.

WHAT IT ENCODES

Triple positivity — simultaneous lupus anticoagulant PLUS anticardiolipin PLUS anti- β 2-glycoprotein-I — marks the highest thrombotic and recurrence risk and mandates indefinite anticoagulation. These patients should be anticoagulated with warfarin (INR 2–3), NOT a DOAC, after the TRAPS trial showed excess arterial events with rivaroxaban in triple-positive patients.

BOARD TRAP

WRONG to equate a single low-titer antibody with triple positivity — risk is graded. A lone, transient, low-titer aCL in an asymptomatic person carries minimal risk, whereas persistent triple positivity is the highest-risk profile.

3. Persistent flag (12 weeks)

WHAT YOU SEE

Small flag 'PERSISTENT +12 wks'.

WHAT IT ENCODES

To count toward APS, an antibody must be confirmed positive on TWO separate occasions AT LEAST 12 WEEKS APART. This interval excludes transient antibodies that commonly appear after infection, acute thrombosis, or inflammation. Persistence is what separates clinically meaningful aPL from an incidental transient finding.

BOARD TRAP

WRONG to confirm APS on a single positive test or on two tests drawn less than 12 weeks apart — a transient post-infectious antibody (e.g., after a viral illness or syphilis) would be falsely labeled APS without the 12-week repeat.

4. Lupus anticoagulant (paradox cannon)

WHAT YOU SEE

'PARADOX CANNON' on the ship's deck.

WHAT IT ENCODES

Lupus anticoagulant is a functional antibody that prolongs phospholipid-dependent clotting assays IN VITRO (prolonged aPTT/dRVVT that fails to correct on mixing study but corrects with added phospholipid) yet causes THROMBOSIS in vivo — the central APS paradox. It is the antibody most strongly associated with thrombotic events of the three criteria antibodies.

BOARD TRAP

WRONG to interpret the prolonged aPTT as a bleeding risk — do NOT transfuse plasma or hold anticoagulation for it. Also note heparin and DOACs cause false-positive LA assays, so test off interfering anticoagulants.

5. Three antibodies / three steps

WHAT YOU SEE

Muscular figure 'five visible body segments' with Step 1-2-3 = LA, aCL, β 2GPI.

WHAT IT ENCODES

The three criteria antibodies are lupus anticoagulant (functional clotting assay), anticardiolipin (ELISA, IgG/IgM), and anti- β 2-glycoprotein-I (ELISA, IgG/IgM). Anti- β 2GPI targets the true pathogenic antigen; IgG isotypes and higher titers correlate best with thrombosis. Order all three when APS is suspected.

BOARD TRAP

WRONG to test only anticardiolipin — the outdated 'aCL only' framing misses lupus anticoagulant (the most thrombogenic) and anti- β 2GPI. Diagnostic workup requires all three antibodies.

6. Phospholipid bead molecules

WHAT YOU SEE

Bead molecules at bottom-left, '1-5%'.

WHAT IT ENCODES

Antiphospholipid antibodies are present in roughly 1–5% of healthy people, with prevalence rising with age and after infections, and they are often transient and low-titer. This background prevalence is exactly why classification demands BOTH a clinical event and persistent, often higher-titer, antibodies.

BOARD TRAP

WRONG to diagnose APS from low-titer aPL in an asymptomatic person — antibody positivity without a qualifying clinical thrombotic or obstetric event is NOT APS. A clinical event is required.

7. Lung-shaped dock (VTE/PE)

WHAT YOU SEE

Lung-shaped harbor with dark clot, 'VENOUS DOCK'.

WHAT IT ENCODES

Venous thromboembolism — deep vein thrombosis and pulmonary embolism — is the most common thrombotic manifestation of APS. Recurrent or unprovoked VTE, VTE at unusual sites, or VTE in a young patient should prompt aPL testing. Lifelong warfarin (INR 2–3) is standard after a first unprovoked VTE in confirmed APS.

BOARD TRAP

WRONG to use a DOAC as first-line for VTE in confirmed APS, particularly triple-positive patients — warfarin to INR 2–3 is preferred because DOACs showed higher recurrence (especially arterial) in trials.

8. Brain-shaped buoy (arterial/stroke)

WHAT YOU SEE

Brain-shaped harbor with blackened/infarcted tissue.

WHAT IT ENCODES

Arterial thrombosis in APS classically presents as ischemic STROKE, often in a young patient without atherosclerotic risk factors; other arterial events include MI and limb/visceral ischemia. After an arterial event, management is warfarin (often target INR 2.5–3.5 or INR 2–3 plus low-dose aspirin) rather than antiplatelet therapy alone.

BOARD TRAP

WRONG to attribute a young patient's stroke to 'cryptogenic' without testing aPL — unexplained stroke in someone young or without atherosclerotic risk factors should prompt an APS workup, and treating with aspirin alone undertreats confirmed APS.

9. Pregnancy dock (obstetric)

WHAT YOU SEE

'HARBOR 4 — PREGNANCY DOCK' with broken baby dolls.

WHAT IT ENCODES

Obstetric APS includes ≥ 3 consecutive early (<10 w) miscarriages, ≥ 1 fetal death at ≥ 10 weeks, or severe pre-eclampsia/placental insufficiency before 34 weeks, driven by placental thrombosis and complement activation. Management in pregnancy is LOW-DOSE ASPIRIN plus prophylactic (or therapeutic) LOW-MOLECULAR-WEIGHT HEPARIN — never warfarin, which is teratogenic.

BOARD TRAP

WRONG to use warfarin in pregnancy — it is teratogenic (warfarin embryopathy) and crosses the placenta. The correct regimen is aspirin + LMWH; warfarin is reserved for the non-pregnant state.

10. Adrenal gland rock (infarction)

WHAT YOU SEE

Adrenal-gland rock with broken clump.

WHAT IT ENCODES

Adrenal vein thrombosis with hemorrhagic infarction can cause primary ADRENAL INSUFFICIENCY in APS. The adrenal venous drainage is prone to thrombosis, and bilateral involvement precipitates an Addisonian crisis with hypotension, hyponatremia, and hyperkalemia.

BOARD TRAP

WRONG to attribute refractory hypotension in an APS patient to sepsis alone — consider acute adrenal insufficiency from adrenal vein thrombosis/infarction; the fix is stress-dose hydrocortisone, not just fluids and pressors.

11. Yellow pipes (atherosclerosis/livedo)

WHAT YOU SEE

'YELLOW PIPES (atherosclerosis)' / lemon-leaks at lower harbor.

WHAT IT ENCODES

APS causes microvascular and cutaneous disease — livedo reticularis/racemosa (a common skin sign, especially with Sneddon syndrome of livedo plus stroke), skin ulcers, and accelerated/premature atherosclerosis from chronic endothelial injury. aPL-nephropathy is a microvascular renal manifestation.

BOARD TRAP

WRONG to dismiss livedo reticularis as benign in a thrombophilic patient — livedo racemosa with strokes (Sneddon syndrome) is strongly aPL-associated and should trigger testing.

12. Thrombocytopenia (massive cannonball)

WHAT YOU SEE

'MASSIVE DARK RED BLOOD CELL CANNONBALL' / low-platelet motif.

WHAT IT ENCODES

Mild-to-moderate thrombocytopenia (typically 50,000–130,000/ μ L) is a common hematologic feature of APS and overlaps with SLE. It is usually not severe enough to cause bleeding and, crucially, does not protect against clotting.

BOARD TRAP

WRONG to withhold anticoagulation because of the low platelets — APS patients clot DESPITE thrombocytopenia, and the dominant clinical problem remains thrombosis, not bleeding (unless platelets are profoundly low).

13. Lighthouse — ≥ 3 clinical / ≥ 3 lab

WHAT YOU SEE

Two lighthouses ' ≥ 3 CLINICAL' and ' ≥ 3 LAB' with classification bottle + clock.

WHAT IT ENCODES

The 2023 ACR/EULAR classification requires an entry positive aPL plus a weighted score of ≥ 3 points from clinical domains AND ≥ 3 points from laboratory domains, with antibodies confirmed persistent over time (the clock). The two highest single items (5 points each) are biopsy-proven microvascular thrombosis and persistent lupus anticoagulant.

BOARD TRAP

WRONG to classify APS from clinical events alone without persistent laboratory positivity — both columns must independently reach 3 points; clinical thrombosis with only transient antibodies does not qualify.

14. Still clots! (refractory)

WHAT YOU SEE

Sign 'STILL CLOTS!' near broken-faucet pipes.

WHAT IT ENCODES

Refractory APS = recurrent thrombosis despite therapeutic anticoagulation. Escalation options: increase warfarin target to INR 3.0–4.0, add low-dose aspirin or hydroxychloroquine, switch heparin/LMWH, and consider statins, rituximab, or eculizumab. Hydroxychloroquine has antithrombotic benefit, especially in SLE-associated APS.

BOARD TRAP

WRONG to assume recurrent clot on warfarin means non-compliance — first CHECK THE INR (subtherapeutic? DOAC switch? drug interactions?). True breakthrough on a therapeutic INR calls for intensifying therapy, not just counseling adherence.

15. Catastrophic APS (CAPS)

WHAT YOU SEE

Storm/escalation zone between the lighthouse and the 'STILL CLOTS!' broken pipes on the right.

WHAT IT ENCODES

Catastrophic APS is the fulminant variant: small-vessel thrombosis striking ≥ 3 organs within ONE WEEK with histologic small-vessel occlusion and aPL positivity, causing rapid multi-organ failure (~1% of APS, ~30–50% mortality). Common triggers are infection, surgery, and abrupt anticoagulation withdrawal. Treatment is the triple bundle: therapeutic anticoagulation + high-dose corticosteroids + plasma exchange and/or IVIG, adding rituximab or eculizumab if refractory.

BOARD TRAP

WRONG to manage CAPS with anticoagulation alone — the rapidly progressive multi-organ thrombosis needs combined anticoagulation + steroids + plasma exchange/IVIG. Treating it like ordinary APS thrombosis is a fatal undertreatment.
